

A random walk in one dimension (with coordinate x) of length N depends on a sequence of N coin tosses. We denote the outcome of the i^{th} coin toss as ω_i with the possibilities $\omega_i = H, T$ corresponding to heads and tails.

Unless mentioned otherwise, the walker always starts at the origin $x = 0$ at time 0. The rule for the random walker is that they move $+1$ units at the i^{th} time step if $\omega_i = H$ and -1 units if $\omega_i = T$. The probability for heads is p and for tails is q with $p + q = 1$. When $p = q = \frac{1}{2}$, this is called a symmetric random walk, and when $p \neq q$, the random walk is called *asymmetric*.

This is described in terms of the random variables $\mathbb{X}_i$, $1 \leq i \leq N$, with each $\mathbb{X}_i$ being a function from the space of coin tosses $\{H, T\}$ to real numbers:

$$\mathbb{X}_i : \{H, T\} \rightarrow \mathbb{R} . \quad (1)$$

The random variable is defined as

$$\mathbb{X}_i(\omega_i) = \begin{cases} +1 & \text{for } \omega_i = H , \\ -1 & \text{for } \omega_i = T . \end{cases} \quad (2)$$

The location of the random walker at the n^{th} time step is also a random variable $\mathbb{M}_n$ which is simply the sum of the first n random variables $\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_n$:

$$\mathbb{M}_n = \sum_{i=1}^n \mathbb{X}_i . \quad (3)$$

Since the right hand side depends on the first n coin tosses, the random variable $\mathbb{M}_n$ depends on all these n coin tosses. We shall however take a more general point of view and consider the $\mathbb{M}_n$ for all $1 \leq n \leq N$ as being functions on the sample space of all possible sequences of N coin tosses Ω_N . A typical sequence in Ω_N is denoted as $\underline{\omega} = \omega_1 \omega_2 \dots \omega_N$ ($\underline{\omega}$ is ω with an underline). Thus, we have

$$\mathbb{M}_n : \Omega_N \rightarrow \mathbb{R} , \quad \text{with} \quad \mathbb{M}_n(\omega_1 \dots \omega_N) = \sum_{i=1}^n \mathbb{X}_i(\omega_i) . \quad (4)$$

Note that even though the definition of $\mathbb{M}_n$ involves all N coin tosses, the actual values that $\mathbb{M}_n$ takes are dependent only on the first n coin tosses.

We studied the following important features of symmetric random walks in class:

1. The probability $p_{n,r}$ of a path of length n leading to (n, k) is

$$\mathbb{P}\{\mathbb{M}_n = k\} = p_{n,k} = \binom{n}{\frac{n+k}{2}} \frac{1}{2^n} . \quad (5)$$

2. The probability $u_{2\nu}$ of return to the origin at time 2ν :

$$\mathbb{P}\{\mathbb{M}_{2\nu} = 0\} = u_{2\nu} = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} . \quad (6)$$

3. The probability $f_{2\nu}$ of first return to the origin at time 2ν :

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} = 0\} = f_{2\nu} = \frac{1}{2\nu-1} \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} . \quad (7)$$

4. The probability of the walker never returning to the origin after 2ν steps:

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2\nu-1} \neq 0, \mathbb{M}_{2\nu} \neq 0\} = u_{2\nu} = \binom{2\nu}{\nu} \frac{1}{2^{2\nu}} . \quad (8)$$

Each of the above probabilities were calculated by counting the number of paths satisfying the appropriate conditions and then multiplying the number by the individual probabilities of the paths occurring. We explore more ideas related to random walks in this assignment.

1 Probability of last visit

So far, we have looked at the probability of return to the origin at time $2n$, as well as the probability of first return to the origin at time $2n$. Now, we shall look at another aspect of random walks which is that of last visit to the origin in a given time duration $2n$.

The random walker starts at the origin $x = 0$ at time 0 as usual and makes their journey based on the outcomes of coin tosses. As time progresses, the random walker returns to the origin occasionally whenever the number of heads and tails in the running count become equal. What we are interested in is one such visit to the origin which is the last in a given time duration $2n$. In other words, after this last visit to the origin (at some time $2k$), the walker does not visit the origin for the remaining time up to $2n$.

More precisely, we ask the following question: given that we track the movement of the random walker from time 0 to time $2n$, what is the probability that the last visit to the origin before $2n$ is at some earlier time $2k$ for some k ?

Let us model this in terms of events involving the location random variables $\mathbb{M}_i$. The situation where walker visits the origin for the last time at time $2k$ in a time period of $2n$ is captured by the event:

$$\mathbb{M}_{2k} = 0 , \quad \mathbb{M}_{2k+1} \neq 0 , \quad \dots , \quad \mathbb{M}_{2n} \neq 0 . \quad (9)$$

In particular, no constraint is placed on the locations of the walker before time $2k$.

Exercise 1a (4 points): The probability of the above event can be captured by splitting the paths into two parts: the first part is from $(0, 0)$ to $(2k, 0)$ with no restrictions as to what happens in between, and the second part from $(2k, 0)$ to $(2n, 2r)$ for some $r \neq 0$ such that the walker never touches or crosses the t -axis.

The two parts are independent of each other since the walker's position is "reset" to the origin at time $2k$. Based on this observation, and the facts we have established previously, show that the probability of last visit at $2k$ is given by

$$\alpha_{2k, 2n} = u_{2k} u_{2n-2k} . \quad (10)$$

You must relate each of the two parts of the path to another counting that we have seen previously. ■

Ans. The probability of return to origin at time $2k$ is u_{2k} (already established in class).

Next, the second part of the path involves starting from the origin at time $2k$ and then going on till time $2n$ while never touching or crossing the t -axis except at the beginning. By a shift of the time variable back by $2k$, this is equivalent to starting at the origin at time $2k - 2k = 0$ and going on till time $2n - 2k$ while never touching or crossing the t -axis. The main lemma discussed in class gives this probability as

$$\mathbb{P}\{\mathbb{M}_1 \neq 0, \mathbb{M}_2 \neq 0, \dots, \mathbb{M}_{2n-2k} \neq 0\} = u_{2n-2k} . \quad (11)$$

The occurrence of the first part of the path from origin to origin in $2k$ time steps is independent of the second part from origin to elsewhere in $2n - 2k$ time steps while never touching or crossing the t -axis. This is because they are governed by non-overlapping parts of the sequence of $2n$ coin tosses. The probability of last visit is then the product of the probabilities of the two parts occurring independently:

$$\alpha_{2k, 2n} = u_{2k} u_{2n-2k} . \quad (12)$$

(1 point for mentioning the probability of return to origin as u_{2k} , 2 points for using the main lemma for the second part of the path, 1 point for using independence of the occurrence of the two parts of the path.)

Exercise 1b (2 points): Plot the probability distribution $\alpha_{2k, 2n}$ for a fixed n and different k in the range $k = 1, 2, \dots, n - 1$. You must generate plots for the following values of n : $n = 5, 10, 20, 40$. ■

Ans. The plots are displayed below in Figure 1.

(Half a point for each plot. I have plotted each plot for k from 0 to n rather than from 1 to $n - 1$ as in the question. Either range for k gets full points.)

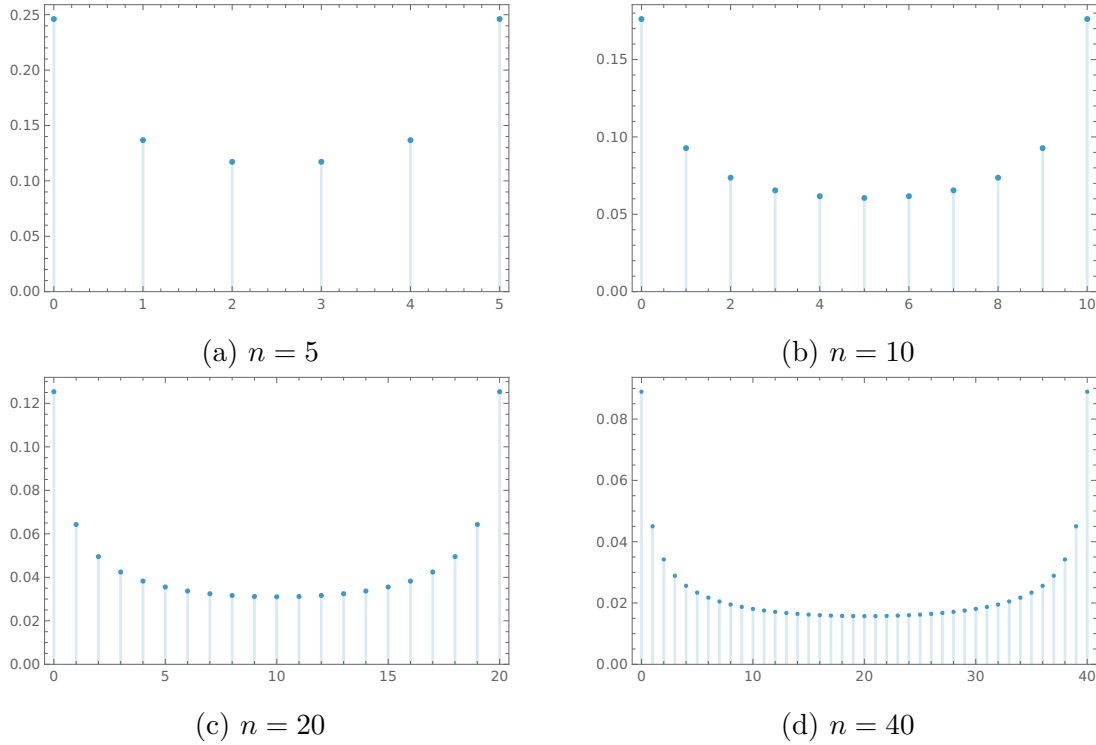


Figure 1: Plots of the last visit probabilities $\alpha_{2k, 2n}$ as a function of k for $n = 5, 10, 20, 40$.

Exercise 1c (2 points): For a given n , for what value(s) of k is the probability of last visit $\alpha_{2k, 2n}$ the smallest, and for what value(s) of k is the probability the largest? You may answer this question by analysing the formula for $\alpha_{2k, 2n}$ or using the plots generated in the previous exercise. ■

Ans. The probability $\alpha_{2k, 2n}$ is smallest for the middle two values $k = \frac{n-1}{2}, \frac{n+1}{2}$ when n is odd, and the unique middle value $k = \frac{n}{2}$ when n is even.

(1 point for the smallest probability and 1 point for the largest probability.)

The last visit probability $\alpha_{2k, 2n}$ is largest for the fringe values of $k = 0$ and $k = n$, corresponding to last visit being at the beginning or the ending of the walk.

Let us look at an application of the above results to a coin-tossing game played by two players for a time duration of $2n$. At any point of time, Player 1 is said to be in the lead if the accumulated number of heads is greater than the accumulated number of tails up to that point in time. Otherwise Player 2 is in the lead.

Relating this to the random walk, it is easy to see that Player 1 is in the lead at a particular time if our random walker is above the t -axis at that time, and Player 2 is in the lead if the random walker is below the t -axis at that time. The two players are tied when the random

walker is at the t -axis, i.e., when the random walker visits the origin.

Till the last visit to the origin at some time $2k$, the lead changes whenever the walker crosses the origin. After the last visit, the lead stops changing since the walker is not allowed to cross the origin again till the game ends at $2n$.

Exercise 1c (2 points): Based on your answers for Exercise 1b, what is the most probable time (or times) when the lead will change for the last time and remain as it is?

Is there anything counter-intuitive in one of the possibilities?

Ans. As explained in the question, the change of lead for the last time corresponds to the last visit to the origin in the time range 0 to $2n$. As we saw above, the highest probability for last visit are at beginning (0) and the end ($2n$) of the walk. Hence, the most probable time for change of lead for the last time is either the beginning or the end.

Most probably, the lead changes at the beginning and remains as it is, or the lead keeps changing till the end. The second possibility is what one usually imagines in such a game – that the lead keeps changing throughout the game.

The first possibility is counter-intuitive since there is a very high probability that the lead changes once near the beginning and stays the same.

(1 point for correctly listing the most probable times of last change of lead and 1 point for correctly identifying the counterintuitive probability along with an explanation as to why it is counterintuitive.)

2 Maxima of random walks

In this exercise, we are interested in the probability of occurrence of paths of length n which start at the origin $(0, 0)$ and never cross the value $x = r \geq 0$ (but may attain $x = r$ at some point along the walk); in other words, we are interested in the probability of occurrence of paths of length n whose maximum position is at most $x = r \geq 0$. In the random variable notation, we would like to compute

$$\mathbb{P}\{\mathbb{M}_1 \leq r, \mathbb{M}_2 \leq r, \dots, \mathbb{M}_n \leq r\} . \quad (13)$$

We split the problem into a series of steps. A typical path of this kind starts at $(0, 0)$ and ends at (n, k) . We need to choose $k \leq r$ since otherwise, any path will cross $x = r$ at some time and the maximum of the path will be $\geq r$. Also, recall that the maximum value k can have is $+n$ for any random path of length n . Thus, if the length of the path $n < r$, then *every* path from $(0, 0)$ to (n, k) has maximum $< r$ automatically and the required probability is 1 in this case. To get interesting answers, we need to consider path lengths n such that $n \geq r$.

So, the ranges of n and k we are interested in are

$$n \geq r, \quad k \leq r. \quad (14)$$

First, we enumerate the number of paths from $(0, 0)$ to (n, k) with $k \leq r$ and such that the paths touch or cross $x = r$ at some time along the walk. This can be achieved by using a version of the reflection principle:

There is a one-to-one correspondence between paths from the origin $(0, 0)$ to (n, k) which touch or cross the $x = r$ line and all paths from $(0, 0)$ to $(n, 2r - k)$, where $(n, 2r - k)$ is the reflection of the point (n, k) across the line $x = r$.

This can be proved by first taking a path from $(0, 0)$ to (n, k) which crosses or touches r and reflecting the portion of the path between the *last* crossing of r and the endpoint (n, k) . This is similar to the proof of the original reflection principle discussed in class where we reflected the portion of the path between the initial point (a, α) and the *first* point where the path touches or crosses the t -axis.

Exercise 2a (2 points): Justify the above reflection principle by choosing some convenient values of r , n and k and drawing a few paths. There is no need to give a formal proof.

■

Ans. Draw a path from $(0, 0)$ to $(n, 2r - k)$ that crosses r , and obtain another path by reflecting the path at the last point of crossing r . This gives a path from $(0, 0)$ to (n, k) . Similarly, start with a path from $(0, 0)$ to (n, k) that crosses r and reflect it at the last point of crossing to get a path from $(0, 0)$ to $(n, 2r - k)$.

(2 points for at least two paths that illustrate the principle.)

Exercise 2b (2 points): Using the reflection principle, show that the probability that a path of length n starts from $(0, 0)$ and ends (n, k) while touching or crossing $x = r$ is

$$\mathbb{P}\{\mathbb{M}_n = 2r - k\} = p_{n, 2r - k} = \binom{n}{\frac{n + 2r - k}{2}} \frac{1}{2^n}. \quad (15)$$

As discussed earlier, the endpoint k must be $\leq r$ because we are ultimately interested in paths which never exceed r even at the endpoints. ■

Ans. The probability of a path from $(0, 0)$ to (n, k) while touching or crossing r is the total number of such paths multiplied by the probability of individual paths $1/2^n$. We argued in Exercise 2a that the number of such paths is equal to the total number of all paths from $(0, 0)$ to $(n, 2r - k)$ which is given by $\binom{n}{\frac{n + 2r - k}{2}}$. Hence,

$$\mathbb{P}\{\mathbb{M}_n = 2r - k\} = \binom{n}{\frac{n + 2r - k}{2}} \frac{1}{2^n}. \quad (16)$$

(1 point for the logic that relates the actual problem to the problem of counting *all paths* from $(0, 0)$ to $(n, 2r - k)$ and 1 point for the final expression.)

Exercise 2c (2 points): Using the answer to the previous exercise, argue that the probability of occurrence of paths from $(0, 0)$ to (n, k) with maximum exactly r is $p_{n, 2r-k} - p_{n, 2(r+1)-k}$. ■

Ans. The paths with maximum value r are obtained by subtracting the number of paths that exceed $r + 1$ from the number of paths that exceed r . This gives the required probability $p_{n, 2r-k} - p_{n, 2(r+1)-k}$.

(2 points if this logic is clearly mentioned.)

Exercise 2d (4 points): Now we finally enumerate all possible paths with maximum equal to r by summing over the allowed values of the endpoint $\mathbb{M}_n = k$. Recall that k must be $\leq r$ since we are interested only in paths which never exceed r . Also, recall that $-n \leq k \leq n$ and that we have restricted n to be $\geq r$. Thus, the upper bound for the sum over k must be r . You can take the lower bound on k to be the minimum possible value of k which is $-n$.

Convince yourself that the above statements are true, and show that the sum over k of the probabilities $(p_{n, 2r-k} - p_{n, 2r+2-k})$ evaluates to

$$\sum_{k=-n}^r (p_{n, 2r-k} - p_{n, 2r+2-k}) = p_{n, r} + p_{n, r+1} . \quad (17)$$

(There are many terms of the form $p_{n, l}$ for some l that is greater than n which you should set to zero. Many terms will be non-zero but most of them will cancel with another term with the opposite sign, finally yielding the answer with only two terms above.)

In fact, the probability of a random walk never exceeding $x = r$ in n time steps is one of $p_{n, r}$ or $p_{n, r+1}$ since either n and r have the same parity or n and $r + 1$ have the same parity:

$$\mathbb{P}\{\mathbb{M}_1 \leq r, \mathbb{M}_2 \leq r, \dots, \mathbb{M}_n \leq r\} = p_{n, r} \text{ or } p_{n, r+1} . \quad (18)$$

■

Ans. As explained in the question, the sum is a telescoping sum. Even though the lower bound in the sum is taken to be $k = -n$, the probabilities are all zero up to the value $k = 2r - n$.

$$\begin{aligned} \sum_{k=-n}^r (p_{n, 2r-k} - p_{n, 2r+2-k}) &= \underbrace{p_{n, 2r+n} - p_{n, 2r+n+2}}_{k=-n} + \underbrace{p_{n, 2r+n-1} - p_{n, 2r+n+1}}_{k=-n+1} + \cdots + \underbrace{p_{n, n} - p_{n, n+2}}_{k=2r-n} \\ &+ \underbrace{p_{n, n-1} - p_{n, n+1}}_{k=2r-n+1} + \underbrace{p_{n, n-2} - p_{n, n}}_{k=2r-n+2} + \underbrace{p_{n, n-3} - p_{n, n-1}}_{k=2r-n+3} + \cdots + \underbrace{p_{n, r+1} - p_{n, r+3}}_{k=r-1} + \underbrace{p_{n, r} - p_{n, r+2}}_{k=r} . \end{aligned} \quad (19)$$

Hence, many terms are zero on their own, or cancel between the successive alternate summands except $p_{n,r}$ in the last summand and $p_{n,r+1}$ in the last but one summand. Since only one of $p_{n,r}$ or $p_{n,r+1}$ is non-zero depending on the parity of n and r , the answer is the appropriate non-zero expression.

(2 points for correctly writing out the telescoping sum and cancelling the intermediate terms + 2 points for the final answer which is one of $p_{n,r}$ or $p_{n,r+1}$.)